Lab 3 - Advanced Machine Learning

Clustering

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General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

K-Means

Type : Partitional
Number of clusters : Specified in advance

Algorithm : Solve *(with $z_i = k \Leftrightarrow i \in \mathcal{C}_k$)*

$$\underset{\mathbf{z} \in [\![1,K]\!]^N}{\text{minimize}} \sum_{k=1}^{K} \sum_{i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad \text{with} \quad \boldsymbol{\mu}_k = \frac{1}{|\mathcal{C}_k|} \sum_{n \in \mathcal{C}_k} \mathbf{x}_n \tag{1}$$

by alternatively assigning points to cluster and recomputing centroids.

Clusters : Compact groups around a centroid

Agglomerative Single Linkage

Type : Hierarchical
Number of clusters : Given by cutting height

Algorithm : Sequentially merge the two nearest clusters and update distance between clusters, until you end up with one cluster

Clusters : Groups connected through chains of nearby points
Robust to : Initialization

DBSCAN

Type : Partitional
Number of clusters : Given by neighborhood radius ε and minimum number of points m

Algorithm :

- Identify core points as having at least m points in their ε-neighborhood
- Add neighbors to the cluster, and re-identify core points
- Non-core points either have been added to a cluster, or are considered as noise

Clusters : Points that are close and in dense regions
Robust to : Initialization and outliers

Illustration of DBSCAN

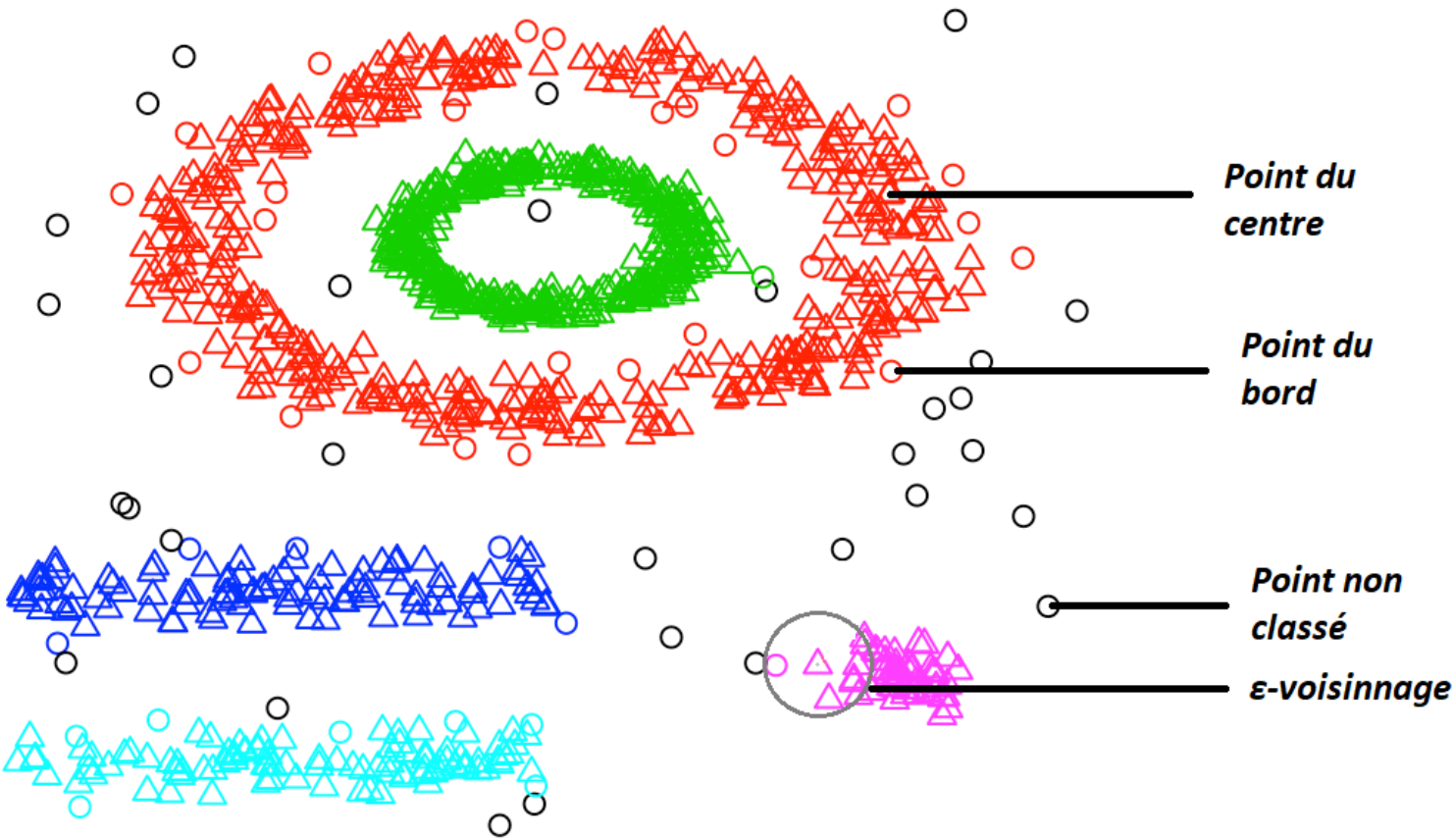


Figure: Clustering results obtained with DBSCAN algorithm.

HDBSCAN

Type : Hierarchical
Number of clusters : Given by minimal density m and minimum cluster size C_{min}

Algorithm :

- Extract the minimum spanning tree of the complete graph, weighted by the mutual reachability distance *(the metric penalizes sparsity)*
- Construct a cluster hierarchy of connected components by removing the heaviest edges *(dense groups emerge as the density threshold increases)*
- Condense the cluster hierarchy based on the minimum cluster size C_{min} *(remove insignificant branches/noise)*
- Extract the most stable clusters from the condensed tree

Clusters : Points that are close and in dense regions that are not easily split
Robust to : Initialization and outliers; no fixed ε required.